have no consciousness, or at most, that they have a very different and inferior type of consciousness.

In order to decide whether animals have conscious minds similar to our own, we must first get a better understanding of how minds function, and what role they play. These are extremely difficult questions, but it is worthwhile to devote some time to them, because the mind will be the hero of several subsequent chapters. We won't be able to grasp the full implications of novel technologies such as artificial intelligence if we don't know what minds are. Hence let's leave aside for a moment the particular question of animal minds, and examine what science knows about minds and consciousness in general. We will focus on examples taken from the study of human consciousness – which is more accessible to us – and later on return to animals and ask whether what's true of humans is also true of our furry and feathery cousins.

To be frank, science knows surprisingly little about mind and consciousness. Current orthodoxy holds that consciousness is created by electrochemical reactions in the brain, and that mental experiences fulfil some essential data-processing function.[3] However, nobody has any idea how a congeries of biochemical reactions and electrical currents in the brain creates the subjective experience of pain, anger or love. Perhaps we will have a solid explanation in ten or fifty years. But as of 2016, we have no such explanation, and we had better be clear about that.

Using fMRI scans, implanted electrodes and other sophisticated gadgets, scientists have certainly identified correlations and even causal links between electrical currents in the brain and various subjective experiences. Just by looking at brain activity, scientists can know whether you are awake, dreaming or in deep sleep. They can briefly flash an image in front of your eyes, just at the threshold of conscious perception, and determine (without asking you) whether you have become aware of the image or not. They have even managed to link individual brain neurons with specific mental content, discovering for example a 'Bill Clinton' neuron and a 'Homer Simpson' neuron. When the 'Bill Clinton' neuron is on, the person is thinking of the forty-second president of the